Quiz #7

1. An email spam filter automatically classifies incoming messages as either spam or legitimate. Messages classified as spam are moved to the spam folder.

   Classify each outcome as a true positive (TP), false positive (FP), true negative (TN), or false negative (FN).

   (a) A legitimate email is moved to the spam folder.

   (b) A spam email is delivered to the inbox.

   (c) A legitimate email is delivered to the inbox.

   (d) A spam email is moved to the spam folder.

2. True or false, and explain:

   (a) The P-value of a test equals its observed significance level.

   (b) The alternative hypothesis is another way of explaining the results; it says the difference is due to chance.

   (c) The variance of the t-distribution curve with 4 degrees of freedom is larger than that of the t-distribution curve with 5 degrees of freedom.

   (d) The t-distribution curve approaches the normal distribution when the degrees of freedom approaches 0.

3. A manufacturer tests light bulbs to determine whether their average lifetime is below the advertised value. A random sample of $n = 100$ bulbs is tested.

   The test has a false-positive rate of $\alpha = 0.05$. For a particular substandard production process, the false-negative rate is $\beta = 0.18$.

   (a) What are the **specificity** and **power** of this test?

   (b) If the manufacturer reduces α while keeping n fixed, what generally happens to β and power?

   (c) How could the manufacturer increase power without increasing α?

4. Suppose that an industrial engineer wishes to test the hypothesis that at most 5% of the products produced in his plant are defective.

   (a) Write down the null and alternative hypothesis.

   (b) The test is:

A. A two tailed test.

B. An upper tailed test.

C. A lower tailed test.

D. Not possible to determine.